CFD Anlaysis of Toroidal Propeller on Shaftless Rim Drive Configuration at Low Advance Ratios

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***Abstract*—This study investigates the effect of a toroidal propeller design on the performance of a rim-driven propulsion system. A comparative analysis was conducted using computational fluid dynamics (CFD) simulations, evaluating the toroidal configuration against a conventional propeller under varying operating conditions at constant intervals. Performance metrics, including thrust, torque, and efficiency, were assessed through graphical representations of key parameters. The results indicate a significant difference between the two designs, with both exhibiting a direct relationship between performance metrics and operating conditions. However, the toroidal configuration demonstrated superior performance across all tested conditions, with the performance gap becoming more pronounced at higher operating speeds.**

Keywords—CFD, Toroidal Propeller, Rim Drive, Shaftless

I. INTRODUCTION

Propeller design is a critical factor in developing efficient propulsion systems. Achieving optimal performance requires careful formulation of parameters along the blade radius, including: Geometric twist (variation in the blade's angle of attack along the radius), Aerodynamic twist (changes in airfoil cross-sectional geometry), and Blade profile optimization (planform shape, chord length, and thickness distribution). Despite these optimizations, inefficiencies persist, primarily due to tip vortices [1]. These vortices arise from pressure differences between the blade's suction and pressure sides, causing turbulent airflow that amplifies drag and reduces lift-to-drag ratios. Recent innovations, such as the toroidal propeller, address these limitations by combining concepts from tandem propellers and contracted-load designs. A tandem propeller employs two counter-rotating blades mounted on a single hub, distributing thrust loads and reducing structural stress compared to conventional designs [2]. This configuration also mitigates induced drag by leveraging the rear blade to extract residual energy from the front blade's wake, improving fluid dynamic efficiency [3]. Meanwhile, the contracted-load propeller minimizes tip vortices by curving the blade tips inward, analogous to winglets on aircraft wings. This design reduces cavitation (formation of vapor bubbles) and delays stall, enhancing efficiency across operating conditions [4]. The toroidal propeller merges these principles into an annular, closed-loop shape. Its continuous blade structure distributes thrust loads evenly and dissipates tip vortices along the span, preventing energy loss from turbulence [5]. For example, the Sharrow Propeller, a commercial marine application, reports 20–30% efficiency gains over conventional propellers due to reduced tip losses and vibration [6]. Similarly, MIT researchers demonstrated toroidal propellers for drones, achieving 50% noise reduction while maintaining thrust [7].

Ducted propulsion systems represent another breakthrough in mitigating turbulence and improving efficiency. These systems enclose the propeller within a shroud or duct, which accelerates incoming airflow and reduces tip vortices by constraining radial flow. The duct's aerodynamic profile generates additional thrust via lift forces along its inner surface, while its structure shields the blade tips from crossflows, further minimizing turbulence [8]. Among recent innovations in ducted propulsion is the hubless configuration, which eliminates the central hub entirely. Conventional hubs create localized turbulence due to their blunt geometry and wake interactions, but hubless designs integrate the propeller blades into the duct's rim, often using magnetic or rim-driven systems. This reduces drag and turbulence caused by hub-induced vortices, further enhancing efficiency [14]. For example, rim-driven thrusters (RDTs) use a ring-shaped motor embedded in the duct rim to rotate the blades, removing the need for a shaft or hub. This design is particularly advantageous in underwater vehicles, where cavitation and mechanical complexity are critical concerns [15]. The duct increases static thrust by up to 30% in hover or slow-speed conditions, making it ideal for underwater drones and VTOL aircraft [9]. By suppressing tip vortices and turbulence, ducted propellers operate more quietly which is critical for urban air mobility and marine applications [10]. The duct protects blades from debris and collisions, extending operational lifespan [11]. However, ducted systems face trade-offs. At high speeds, duct drag can offset efficiency gains, and their weight complicates integration into lightweight platforms [12]. Recent advances, such as adaptive ducts with variable geometry and hubless rim-driven systems, aim to balance performance across varying freestream velocity [13, 16].

A. Statement of the Problem

Rim driven Propulsion proves itself to be efficient when compared to hub propulsion based upon the indicated papers. Meanwhile toroidal blade design also contributes to the overall performance of the system as they help mitigate tip vortices. These discoveries and findings of previous literatures provides

a significant insight on improving the design of the propulsion system as a whole.

However, literature with regards to the implementation of both in a single system is lacking. The author would like to implement both configurations to come up with a more efficient design capable of performing better at all operating condition. This involves comparing the existing design which a conventional blade which is also rim driven. The goal of this paper is to:

- To quantify the performance variables of both propeller configuration such as Thrust, Torque, efficiency at various Advance ratio. This can be done using CFD simulation allowing reliable data to be gathered.
- Compare the Torque Thrust and efficiency with respect to the changing advance ratio with both conventional and toroidal design.
- To Analyze the behavior of velocity profile and their behavior along the surface of the blade in order to justify the quantified data extracted from the simulation.

II. Methodology

A. Geometry Preperation

This configuration is based upon the parameters that are optimized by previous literatures. Meanwhile for the conventional Configuration, the parameters are described by:

TABLE I. GEOMETRIC PARAMETERS

Parameters	Values
Tip Ratio	0.4
Chord Length	60 mm
Blade Length	135 mm
Center Space	15 *mm*
Airfoil	NACA 2412
Blade Diameter	437.34 mm

[a] These configurations are applied on both propeller



Fig. 1. Toroidal Propeller

The specification above refers accordingly to a marine propeller's typical operating range having 100,000 to 1,000,000 Reynolds number.

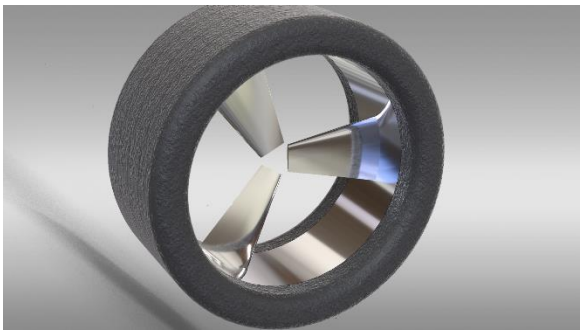


Fig. 2. Toroidal Propeller

The specification above refers accordingly to a marine propeller's typical operating range having 100,000 to 1,000,000 Reynolds number. The Airfoil NACA 2412 has existing data with regards of this range therefore by utilizing the data we could calculate the approximate blade twist while also taking account for the propeller's taper ratio. The formula used for calculating the Blade twist would be:

$$\theta_{tip} = \theta_{root} + \theta_{root}(Taper\ Ratio) \quad (1)$$

Meanwhile on the case of toroidal propeller a positive stagger is applied similar to a tandem propeller configuration while retaining the constant geometric specification of the conventional rim drive configuration.

B. Meshing

Meshing is the process of discretizing the computational domain to a particular size and number of cells. It is crucial as it paves way for us to conduct steps that may greatly affect the accuracy of our data. In this study a meshing scheme is done using Hex dominant algorithm of SIMSCALE with moderate fineness for automated sizing. Due to the limited computational capacity the maximum mesh time was restricted to 5 Hours which results to 1.1 million cells and 1.4 Million nodes to be generated shown on figure 3. The computational domain is cylindrical having radius of 10D and a height of 3D upstream and 7D downstream where D is the diameter of the propeller.

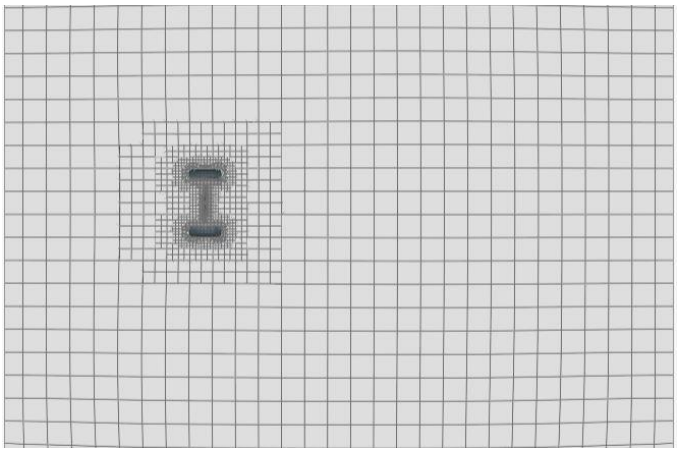


Fig. 3. Mesh Configuration

C. Performance Variables

This research focuses on providing insights with the performance of two propeller configurations: the toroidal propeller and the conventional one, both equipped with rim drive mechanisms. The Parameters which influence the performance of the propeller are defined by the following equations:

$$J = \frac{V}{nD} \tag{2}$$

The Advance ratio expressed as J is the ratio of the propeller's relative velocity V to the rotational velocity n multiplied by the propeller's diameter D is very essential in analyzing propeller's performance. It tells us how the propeller behaves at various operating conditions, thus making it crucial as a parameter.

$$\eta = \frac{C_T J}{C_Q 2\pi} \tag{3}$$

Another Parameter of propeller's performance is the Efficiency denoted by $\boldsymbol{\eta}$**.** It is the ratio of the Power Output which is $\boldsymbol{C_T}$for thrust and J for the advance coefficient over C_p which is the energy required for the change of kinetic energy from the front of the propeller to its rear. This tells us how much input power is converted into output by the propeller.

$$C_{\mathrm{T}} = \frac{T}{\rho n^2 D^4} \tag{4}$$

The coefficient of thrust expressed as C_T pertains to the dimensionless constant that defines the Parameter affecting the force on the propeller's axial direction.

$$C_{\mathrm{Q}} = \frac{Q}{\rho n^2 D^5} \tag{5}$$

Meanwhile the coefficient of torque expressed as C_{Q} in equation 5 depicts the dimensionless constant that defines the work acting upon the radial direction of the propeller.

D. Operating Conditions

Advance ratio is a crucial factor when modeling boundary condition such as freestream velocity and angular velocity. In this case, we are going to investigate the changing values of performance variable by making the angular velocity a function of advance ratio while freestream velocity is constant. Typical cruising speed for small scale UAV is 80 km/h or 22.22 m/while for the advance ratio, due to computational constraints, low values of advance ratio is implemented. The values of angular velocity in RPM are indicated on table below. Brushless motor typical RPM rating has 10000 RPM thus this assumed value of angular velocity makes sense. The operating condition is assumed to be incompressible since it does exceed values with that of compressible.

Advance Ratio (J)	Freestream Velocity	Angular Velocity
0.1	3.6445 m/s	5000 RPM
0.2	7.289 m/s	5000 RPM
0.3	10.9335 m/s	5000 RPM

TABLE II. OPERATING CONDITIONS

E. Simulation Scheme

This study employs steady-state analysis to evaluate the propeller's performance, obtaining key performance parameters through CFD simulations. The analysis is conducted using the Finite Volume Method (FVM) and is based on the Reynolds-Averaged Navier-Stokes (RANS) equations, expressed as follows.

$$\frac{\partial \overline{u_i}}{\partial t} + \overline{u_j}\frac{\partial \overline{u_i}}{\partial x_j} = -\frac{1}{\rho}\frac{\partial \bar{p}}{\partial x_i} + \nu\frac{\partial^2 \overline{u_i}}{\partial x_j \partial x_j} - \frac{\partial \tau_{ij}}{\partial x_j} \tag{6}$$

Since the fluctuating component which is the Reynold's Stress Tensor $\tau_{ij} = \overline{u'_i u'_i}$ is unknown, we approach this using Bousinessq hypothesis to express the term from the known value:

$$u'_i u'_i = -\mu_t\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right) + \frac{2}{3}\rho k \delta_{ij} \tag{7}$$

The turbulent viscosity is then treated using k-SST omega turbulence model.

III. RESULTS AND DISCUSSION

The analysis reveals distinct trends in thrust, torque, and efficiency between the conventional and toroidal propeller configurations. The toroidal configuration exhibits improved thrust performance at higher advance ratios, surpassing the conventional counterpart as freestream velocity increases. Regarding torque, the toroidal configuration demonstrates a lower value at low freestream velocity, indicative of its efficient force distribution along the propeller surface. This efficiency in load distribution directly contributes to the reduction of torque and aligns with findings from previous studies. Additionally, the toroidal configuration maintains competitive efficiency across varying operating conditions, showcasing its potential for enhanced aerodynamic performance and reduced energy losses compared to the conventional design. The following graphs depicts the dynamic relationship of these variables with respect to advance ratio.

A. Thrust

Thrust is a critical parameter in assessing the performance of a propeller, as it represents the total axial force generated. Figure 4 illustrates the relationship between thrust and the advance ratio. It is evident that the conventional configuration surpasses the toroidal configuration at an advance ratio of 0.1. However, at operating conditions with higher freestream velocities,

The following figure shows the relationship of each configuration with regards to thrust.

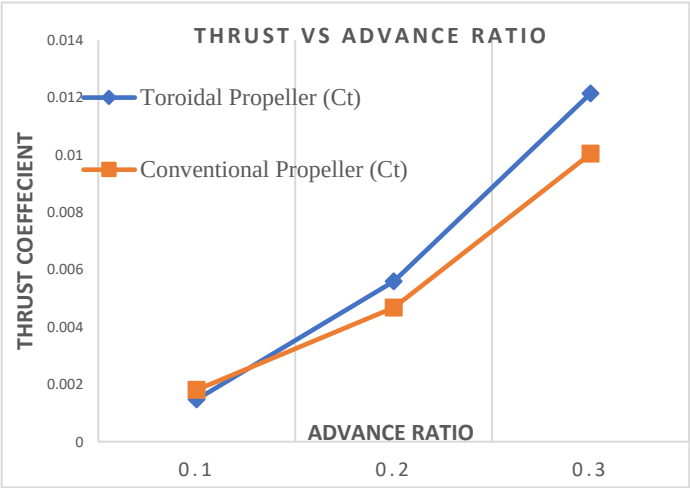


Fig. 4. Thrust at varying advance ratio

B. *Torque*

The torque characteristics of the conventional and toroidal configurations show a significant disparity, particularly at a 0.1 angle of attack, where the conventional configuration exhibits a higher torque value. Torque is calculated as the sum of the normal forces and wall shear stress distributed along the surface of the propeller relative to its radius. This observation suggests that the toroidal configuration achieves a force distribution that considerably reduces torque. Furthermore, this finding aligns with previous literature, which indicates that the toroidal configuration enables a more efficient load distribution along its span.

Meanwhile, as the advance ratio increases, the torque values of the toroidal configuration gradually converge with those of the conventional configuration. This indicates that at higher advance ratios, the disparity in torque diminishes, suggesting comparable performance between the two configurations under these operating conditions.

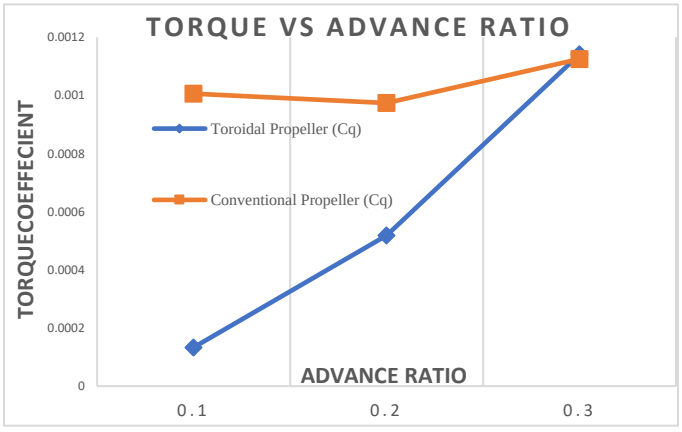


Fig. 5. Torque at varying advance ratio

C. *Effeciency*

The Efficiency Graph shown on figure 6 shows a significant gap between the two configuration even at various operating The efficiency graph depicted in Figure 6 highlights a notable disparity between the conventional and toroidal configurations across a range of operating conditions. This gap is primarily attributed to the dynamic interplay of torque and thrust values as the advance ratio varies. The toroidal configuration consistently demonstrates dominance in efficiency, largely due to its optimized aerodynamic design, which facilitates a more uniform distribution of forces along the propeller surface. By achieving a lower torque value and maintaining higher thrust under diverse conditions, the toroidal configuration effectively minimizes energy losses while maximizing propeller performance. This superior efficiency underscores its potential advantages in applications requiring enhanced aerodynamic stability and energy conservation.

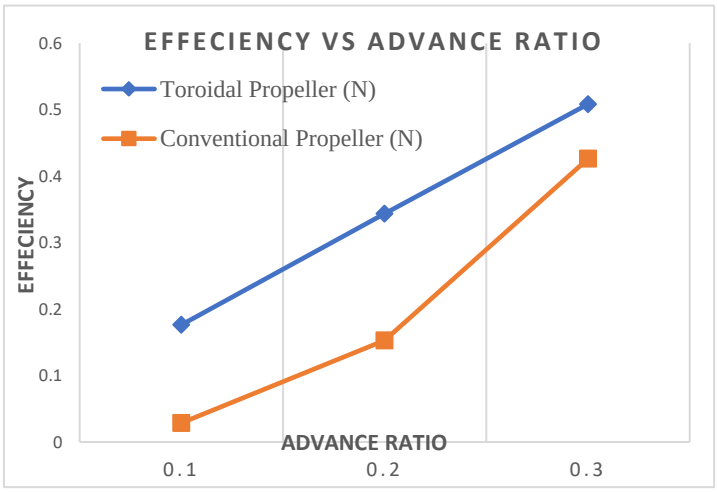


Fig. 6. Effeciency at varying advance ratio

D. *Velocity Profile*

In order to validate the quantitative findings, velocity profiles for both configurations were analyzed. Figures 7 and 8 illustrate the airflow characteristics at an advance ratio of 0.3.

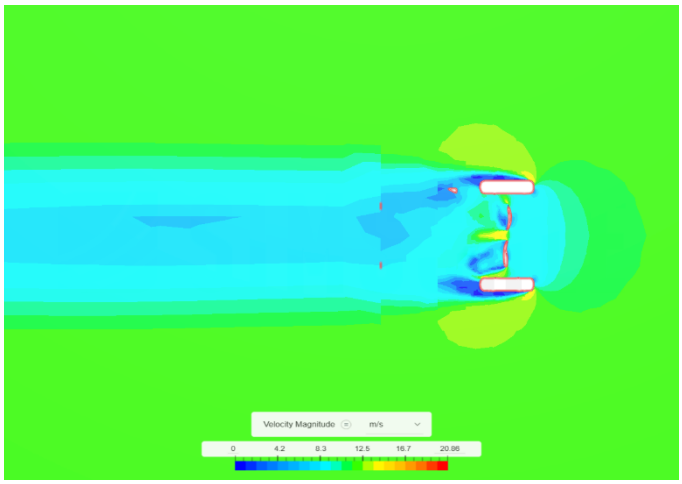


Fig. 7. Velocity Profile of Toroidal Cofiguration.

While both configurations share similarities in their velocity contours, it is apparent that the toroidal configuration exhibits a slightly superior ability to sustain smooth airflow compared to the conventional design. Notably, in the region between the blades, the toroidal configuration demonstrates efficient dispersion of airflow, reducing turbulence and ensuring a more stable flow pattern. This behavior highlights the aerodynamic advantage of the toroidal design, as it minimizes energy losses and enhances overall performance. Furthermore, the smoother velocity profile correlates with the reduced torque and improved efficiency observed in prior analyses, reinforcing the proposition that the toroidal configuration offers significant improvements in load distribution and aerodynamic stability.

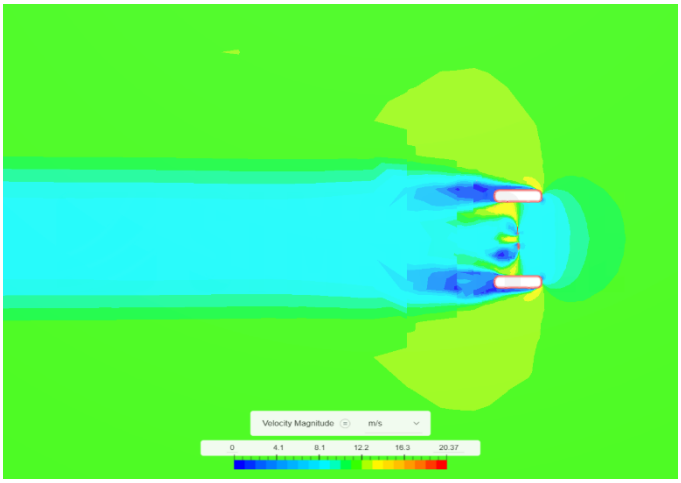


Fig. 8. Velocity profile of Conventional Configuration

RECOMMENDATIONS

This study provides crucial insights with regards of the relationship between parameters affecting Rim Driven Propulsion performance, however there are still some gaps which the author recommends for future researchers:

- The methodology used for this study which is steady state analysis cannot provide accurate insight with variables that are unsteady such as vortex shedding and cavitation, thus using an approach where these variables are accurately documented like transient analysis is crucial to come up with a reliable finding.
- The duct design used in the study is a simplification used only to captivate the change focusing on the propeller's design, therefore various duct design (which is the rim) such that of Kort nozzle are highly suggested to establish higher efficiency.
- Both propeller's geometric parameters such as twist, blade angle, rake, center gaps, number of blades, undulation and serration, and shape of the blades should be further more investigated on how these changes influence the overall performance of the system and its synergy on implementing various duct design.
- The mesh method used in the study relies only to the default setting of the solver and has only a limited number of cells due to computational cost, therefore mesh independence study should be conducted along with time dependence study when using transient analysis to obtain further accuracy in gathering data.
- The solver used in this study is not versatile when it comes to the approach in solving for the fluctuating component of the RANS approach thus, using solver such as ANSYS and Open Foam is highly recommended.
- For verification process, an actual 3d model of the CAD designs should be used in order to realistically capture the data of parameters involved especially the advance ratio.

Although this study shows the potential of the Toroidal configuration as a pivotal variable in enhancing the performance of propeller, the gaps mentioned should also be taken into account in order to come up with an accurate and reliable result.

ACKNOWLEDGMENT

The author wishes to express profound gratitude to all those who have provided invaluable support—be it financial, emotional, or intellectual—throughout the course of this endeavor. Special thanks are extended to my instructor, whose generous understanding and guidance allowed me the time and opportunity to bring this research to fruition. This work would not have been possible without their encouragement. I am deeply indebted to my parents for their unwavering support and for funding my education, as well as to my uncle for his generous sponsorship of the laptop that made this research feasible. Above all, I thank God for blessing me with such a supportive and loving family, without whom my dreams would remain unattainable.

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